Image Captioning

Problem Description :

Dataset Link : https://www.kaggle.com/shadabhussain/flickr8k.

Project Overview

Image captioning is a fascinating application of computer vision and natural language processing (NLP) that involves generating textual descriptions for images. The aim of this project is to build a deep learning model that can automatically generate captions for images. The project involves working with image data, natural language processing techniques, and building deep learning models.

Dataset

For this project, we will use the Flickr8k dataset, which contains 8,000 images, each with five captions. The dataset is available on Kaggle and can be downloaded using the following link: https://www.kaggle.com/shadabhussain/flickr8k.

Project Objectives

The main objectives of the project are:

1. Preprocessing the image and text data to prepare it for use in a deep learning model
2. Building a deep learning model to generate image captions
3. Training and optimizing the model to improve its performance
4. Evaluating the performance of the model using metrics such as BLEU score, METEOR score, and ROUGE score

Deliverables

The project deliverables are:

1. A deep learning model that can generate captions for images

2. A report outlining the approach taken to build the model, including the preprocessing steps, the architecture of the model, the optimization techniques used, and the evaluation results
3. A Python script that can generate captions for new images

Possible Framework :

1. **Data preprocessing**: This involves loading and preparing the image and text data for use in a deep learning model. The image data can be preprocessed using techniques such as resizing, cropping, and normalization. The text data can be preprocessed using techniques such as tokenization, stop word removal, and stemming.
2. **Model architecture**: This involves selecting an appropriate deep learning model architecture for generating image captions. The model can be based on either a CNN-RNN architecture or a transformer-based architecture.
3. **Training**: This involves training the deep learning model using the preprocessed image and text data. The model can be trained using techniques such as gradient descent, stochastic gradient descent, and backpropagation.
4. **Optimization**: This involves optimizing the deep learning model to improve its performance. This can be done using techniques such as regularization, dropout, and batch normalization.
5. **Evaluation**: This involves evaluating the performance of the deep learning model using metrics such as BLEU score, METEOR score, and ROUGE score.

Code Explanation :

Here is the simple explanation for the code which is provided in the code.py file.

The code starts by importing the necessary packages, including TensorFlow, NumPy, Pandas, os, and more.

Next, we define a function to preprocess the image data. This involves resizing the images, normalizing the pixel values, and using the InceptionV3 pre-trained model to extract image features.

We then define a function to preprocess the text data. This involves removing punctuation, converting all text to lowercase, and creating a vocabulary of words in the text.

We load the dataset and split it into training and validation sets. We preprocess the data using the functions defined earlier.

Next, we define the model architecture. This includes the image model and the text model. We then combine the two models and train the model using the training data.

After training the model, we evaluate it using the validation data and generate captions for new images using the trained model.

Overall, the code uses a combination of image and text data to generate captions for images. It involves pre-processing the data, defining the model architecture, and training and evaluating the model.

Documentation :

Image Captioning

This project aims to generate a natural language description of an image. The model takes an input image and generates a caption that describes the contents of the image.

Dataset

The dataset used for this project is the Flickr8k dataset, which contains over 8000 images and associated captions. Each image has 5 captions describing it, making it a total of 40,000 captions.

Project Overview

The project can be broken down into the following steps:

1. Load and preprocess the image and text data
2. Extract features from the image using a pre-trained CNN
3. Preprocess the captions by tokenizing and encoding them
4. Define and train the model
5. Evaluate the model on test data and generate captions for new images

Required Libraries

The following libraries are required to run this project:

- numpy
- pandas
- matplotlib
- PIL
- tensorflow

Running the Project

1. Download the Flickr8k dataset from the Kaggle link provided
2. Unzip the dataset and place it in a folder named **Flickr8k** in the project directory
3. Run the **preprocess_data.py** file to preprocess the image and caption data
4. Run the **train_model.py** file to train the model

5. Run the **evaluate_model.py** file to evaluate the model on test data and generate captions for new images

Future Improvements

- Fine-tune the pre-trained CNN for better feature extraction
- Implement a more advanced language model such as an LSTM or Transformer
- Use a larger dataset such as Flickr30k or COCO for better performance

References

- Dataset: https://www.kaggle.com/adityajn105/flickr8k
- Paper: "Show and Tell: A Neural Image Caption Generator" by Vinyals et al. (2015)

Exercise :

Try to answers the following questions by yourself to check your understanding for this project. If stuck, detailed answers for the questions are also provided.

Here.